

Yi-Chun Liao

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RESEARCH INTERESTS

Focus on the intersection of computer architecture and machine learning, especially on co-designing hardware and software to enhance resource efficiency and scalability of domain-specific memory-centric accelerators.

EDUCATION

National Taiwan University (NTU)

B.Sc. in Computer Science and Information Engineering (CSIE)

- Overall GPA: 3.90/4.30, Last 60 GPA: 4.09/4.30
- Deferred graduation for 1 semester due to compulsory military service.

Taipei, Taiwan

Sep. 2021 – Dec. 2025

PUBLICATIONS

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “RETENTION: Resource-Efficient Tree-Based Ensemble Model Acceleration with Content-Addressable Memory,” Under review by *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, doi: [10.48550/arXiv.2506.05994](https://doi.org/10.48550/arXiv.2506.05994)

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “Practicalizing Tree-Based Model Acceleration with CAM through Model Pruning and Data Placement Optimization.” *ACM/IEEE International Conference on Hardware/Software Codesign and System Synthesis (CODES+ISSS '25)*, September 28-October 3, 2025, Taipei, Taiwan. ACM, New York, NY, USA, 2 pages. doi:[10.1145/3742873.3758342](https://doi.org/10.1145/3742873.3758342)

Hong-Wei Li, Ping-Ting Liu, Bo-Wei Lin, **Yi-Chun Liao**, and Yennun Huang. “IPMES: A Tool for Incremental TTP Detection Over the System Audit Event Stream,” *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Brisbane, Australia, 2024, pp. 265-273, doi:[10.1109/DSN58291.2024.00036](https://doi.org/10.1109/DSN58291.2024.00036)

RESEARCH EXPERIENCE

Research Intern

University of Notre Dame, Advisor: [Prof. Xiaobo Sharon Hu](#)

Indiana, United States

May 2025 – Present

- Conduct independent research on realizing in-memory retrieval for RAG, leveraging a quantization scheme to meet hardware constraints and proposing an algorithm to introduce new capability for content-addressable memory (CAM).
- Conduct independent research on alleviating the scalability issue of in-memory hyperdimensional computing, redesigning model to optimize performance under hardware constraints and introducing an algorithm to enhance CAM scalability.
- Contribute to the extension of open-source CAM simulator [CAMASim](#), focusing on data placement optimization with the approaches proposed in the first-authored paper RETENTION.

Research Assistant

Academia Sinica, Advisors: [Prof. Tei-Wei Kuo](#) and [Prof. Yuan-Hao Chang](#)

Taipei, Taiwan

Apr. 2024 – Present

- Conduct independent research on minimizing resource consumption of in-memory inference for tree-based model, introducing an end-to-end framework [RETENTION](#) to enhance resource efficiency for CAM-based acceleration.
- Explore and quantify the relationship between capacity redundancy and processing overhead, enabling tradeoffs between controllable accuracy loss, energy consumption, and capacity requirement.
- Propose a pruning algorithm to minimize model complexity within controllable accuracy loss, and two data placement strategies to prioritize space and energy efficiency respectively.
- Improve space efficiency by $4.35\times$ to $207.12\times$ with less than 3% accuracy loss and no degradation on throughput and energy efficiency. The first-authored paper is currently under review by TCAD.
- Collaborate on minimizing retrieved tokens to accelerate RAG, exploring the tradeoffs between accuracy, chunk size, and capacity requirement.

Research Assistant

Academia Sinica, Advisor: [Prof. Yennun Huang](#)

Taipei, Taiwan

Jul. 2023 – Jun. 2024

- Collaborated on developing intrusion detection tool [IPMES](#) to detect incremental tactics, techniques, and procedures (TTPs) in system audit event streams. The co-authored paper was published in DSN 2024.
- Constructed an automated system independently to record hundreds of system attacks, utilizing [MITRE Caldera](#) to inject malware and TTPs into virtual machines. The system was subsequently used for synthesizing system attack datasets.

HONORS

First Place of 2025 Bachelor Thesis Award <i>National Taiwan University</i> Distinguished through a three-stage selection process and recognized as one of the top 2 in the CSIE Department, the best in the EECS College, and one of the top 6 university-wide.	Taipei, Taiwan <i>Jun. 2025</i>
Appier Enterprise Award <i>National Taiwan University</i> Received the Most Potential Award from Appier in the departmental undergraduate research exhibition.	Taipei, Taiwan <i>Jun. 2025</i>
Dean's List Award <i>National Taiwan University</i> Ranked 1/155 in the department for both Spring and Fall 2024 semesters.	Taipei, Taiwan <i>Mar. 2025</i>
Irving T. Ho Memorial Scholarship <i>National Taiwan University</i> Awarded to 3 students in the department for academic and research excellence.	Taipei, Taiwan <i>Feb. 2025</i>
First Place of AUO Enterprise Award <i>MakeNTU 2024 (Nationalwide Makerthon)</i> Designed an interactive in-car RPG game for AUO smart windows to enhance travel experience, which was recognized as the best work for AUO Enterprise Award among 43 teams.	Taipei, Taiwan <i>May 2024</i>

TEACHING EXPERIENCE

Teaching Assistant <i>Computer Architecture, Lecturer: Prof. Chia-Lin Yang</i> - Supported the instruction of the junior compulsory course Computer Architecture at NTU, enrolled by over 100 students. - Led weekly discussion sessions, designed and/or graded assignments, quizzes, and exams. - Provided voluntary office hours, exceeding 10 hours per week during exam periods.	Taipei, Taiwan <i>Sep. 2024 – Jan. 2025</i>
Teaching Assistant <i>APCS Camp</i> - Provided TA hours for over 300 C++ beginners during two-week crash courses. - Adapted teaching strategies including programming problem solving and customized course reviews to meet the diverse needs and skill levels of students.	Taipei, Taiwan <i>Jul. 2022/23 – Aug. 2022/23</i>

EXTRACURRICULAR ACTIVITIES

President <i>NTU CSIE Department Student Association</i> - Directed the student council of the CSIE department with over 50 members. - Promoted student welfare and represented over 600 students in faculty meetings, contributing to departmental decisions.	Taipei, Taiwan <i>Aug. 2023 – Jul. 2024</i>
Vice President <i>NTU EECS College Student Association</i> Represented over 1500 students in college faculty meetings and helped shape student-related policies.	Taipei, Taiwan <i>Aug. 2023 – Jul. 2024</i>
President <i>NTU Cancer Children Service Club</i> - Led a 50-member service club, managing budget allocation and arranging educational and recreational activities. - Worked with Childhood Cancer Foundation of R.O.C., organizing voluntary services to reduce the burden of parents.	Taipei, Taiwan <i>Feb. 2023 – Jul. 2023</i>
Vice Coordinator <i>NTU CSIE Camp</i> - Organized a 6-day camp for over 100 high school students with a 50-member staff team. - Designed activities to introduce students to the CSIE department and provide insights into university life.	Taipei, Taiwan <i>Feb. 2023 – Jul. 2023</i>

SKILLS

- Language Proficiency:** English (Proficient, TOEFL iBT 112/120 with Speaking 28/30), Mandarin (Native)
- Technical Skills:** C, C++, Python (PyTorch, NumPy, scikit-learn), Java, JavaScript, LaTeX, Linux, Git